

Multivariable Calculus (MT1008)

Sessional-II

Date: 08th April, 2026

Total Time: 1 Hours

Course Instructor(s)

Total Marks: 65

Dr. Akhlaq Ahmad Bhatti (Moderator)
Dr. Mazhar Hussain, Dr. Nasir Ali,
Dr. Aziz, Dr. Imran Nadeem,
Dr. Hina Dilawar, Ms. Laila Yawar,
Ms. Eesha Mir

Total Questions: 5

Section

Student Signature

Roll No

Important Instruction: 1. Only simple calculator is allowed. Use of programmable calculator or any helpful devices will result F grade in the course.

2. Question # 1(f) must be returned to the invigilator along with answer sheet.

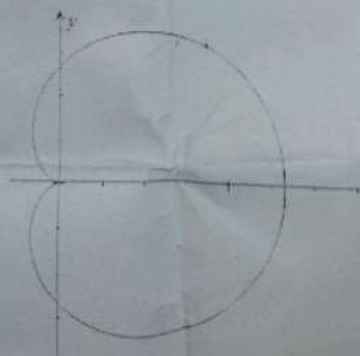
CLO #2: Evaluation of Multiple Integrals in Different Coordinate Systems and Their Applications to Work, Circulation, Flux, Green's Theorem, and Stokes' Theorem.

Question # 1(a)[10]: Set up (ONLY) the below Cartesian integral into an equivalent Polar integral.

$$\int_{x=-1}^1 \int_{y=-\sqrt{1-x^2}}^{\sqrt{1-x^2}} \frac{2}{(1+x^2+y^2)^2} dy dx.$$

Question # 1(b)[10]: Find the area of the region (sketch the region on answer sheet) that lies inside the cardioid $r = 1 + \cos\theta$ (polar graph given below) and outside the circle $r = 1$.

polar coordinates.



$$\cos 2\theta = \cos^2\theta - \sin^2\theta$$

Question # 1(c)[10]: Use the transformation $u = x - y, v = 2x + y$ to evaluate the integral

$$\iint_R (2x^2 - xy - y^2) dx dy,$$

for the region R in the first quadrant bounded by the lines $y = -2x + 4$, $y = -2x + 7$, $y = x - 2$, and $y = x + 1$.

Question # 1(d)[10]: Set up (ONLY) triple integral for the volume of the sphere $\rho = 2$ in (a) spherical, (b) cylindrical coordinates.

Question # 1(e)[10]: Find the line integral of $f(x, y) = ye^{x^2}$ along the curve

$$\mathbf{r}(t) = 4t \mathbf{i} - 3t \mathbf{j}, 1 \leq t \leq 2.$$

$$2 \left(\frac{5V+V}{3} \right)^2 - \left(\frac{5V+V}{3} \right) \left(\frac{V-2V}{3} \right) - \left(\frac{V-2V}{3} \right)^2$$

$$\frac{2}{9} \left((5V)^2 + 2(5V)(V) + V^2 \right) - \frac{1}{9} (5V(V-2V) + V(V-2V)) - \frac{1}{9} (V)^2 - 2(V)(-2V)$$

$$\frac{2}{9} (25V^2 + 10VV + V^2) - \frac{1}{9} (5VV - 10V + V^2 - 2VV) - \frac{1}{9} (V^2 - 4VV + 4V^2)$$

$$\frac{2}{9} (50V^2 + 20VV + 2V^2) - \frac{1}{9} (3VV + 10V - V^2 + 4VV + 4V^2)$$

$$\frac{2}{9} (50V^2 + 24VV + 10V)$$

$$y = \frac{V-2V}{3} \quad x = \frac{5V+V}{3}$$

$$r = \sqrt{x^2 + y^2}$$

$$\begin{aligned}\theta &= r \sin \phi \\ z &= r \cos \phi \\ r &= \rho \sin \theta \\ x &= r \cos \theta\end{aligned}$$